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Scalability-Aware Benchmarking of Graph Anomaly Detection on Financial and Cyber Networks: Evaluated Population, Alert Budgets, and the Exact Limits of Decoupled GNNs

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Abstract

Graph anomaly detection (GAD) is increasingly adopted for financial fraud mitigation and cyber intrusion defense. However, real-world deployment faces severe obstacles: out-of-memory (OOM) failures on massive graphs are often obscured by selective reporting, severe class imbalance invalidates plain ROC-AUC rankings, and unsupervised reconstruction objectives can paradoxically assign higher anomaly scores to licit hub entities than to simple fraudulent chains. To establish a rigorous, application-oriented benchmark, we evaluate seven functioning primary detectors (DOMINANT, AnomalyDAE, CoLA, GADNR, OCGNN, DLG-Base, and DLG-Aug) across 13 primary graph benchmarks (five native financial, eight synthetic injected) and one external enterprise cybersecurity network (LANL-RedTeam) over five random seeds (455 total runs). We enforce a strict support-aware contract: 84 of 91 primary model–dataset pairs are successfully completed, while unsupported cells are reported transparently rather than silently imputed. Our empirical results reveal critical domain-dependent behaviors. While Decoupled Local-to-Global augmentation (DLG-Aug) improves average precision (AP) over the DLG-Base reference on the Elliptic transaction network by $+0.0350$ (0.1037 vs. 0.0687), it suffers an AP degradation of -0.0656 on the large Reddit-Syn graph (0.0573 vs. 0.1229) and exhibits negative delta on LANL-RedTeam. On blockchain token-transfer networks, at least one established comparator exceeds DLG-Aug in mean AP on each of the three blockchain datasets (Ethereum: DOMINANT 0.7410 vs. Aug 0.6976; BSC: DOMINANT 0.6657 vs. Aug 0.6276; Polygon: OCGNN 0.4851 vs. Aug 0.4578). Furthermore, we audit the phenomenon of below-random ROC-AUC values (< 0.50) observed on Elliptic across reconstruction baselines, proving through exhaustive label, score, and mask verification that this stems from normal hub reconstruction residuals rather than software defects. Finally, we provide closed-form exact Gram decoders that eliminate dense $O(N^2)$ memory allocation and present a practical decision matrix connecting alert budgets (top-1% and top-5%) to operational detector selection.

Keywords: graph anomaly detection; graph neural networks; financial fraud detection; alert budget; exact sparse reconstruction; reproducible benchmarking; local-to-global learning

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1. Introduction

Graph anomaly detection (GAD) has emerged as a cornerstone of modern machine learning for financial fraud investigation, blockchain anti-money laundering (AML), and

enterprise cybersecurity monitoring [1,2]. By modeling transactions, contracts, and authentication events as interconnected graphs $G = (V, E, X)$, graph neural networks (GNNs) can capture subtle topological irregularities and feature discrepancies that evade tabular screening.

However, moving GAD from idealized academic benchmarks to operational deployment reveals fundamental challenges that conventional evaluation protocols frequently obscure:

1. **Failure-Blind Reporting and Hardware Infeasibility:** Standard leaderboards often omit unsupported cells when models suffer out-of-memory (OOM) crashes or runtime timeouts on web-scale or multi-million-node graphs, or they quietly replace the full-graph objective with localized neighborhood sampling without verifying whether score semantics are preserved.
2. **Imbalance Disconnect and Below-Random Inversion:** In real-world financial fraud, anomaly prevalence is frequently below 2% or 0.5%. In these extreme regimes, high area under the ROC curve (ROC-AUC) can be misleading. More critically, unsupervised reconstruction models (e.g., autoencoders reconstructing node attributes and adjacency) often assign their highest reconstruction errors to high-degree, structurally complex licit hubs (such as cryptocurrency exchanges or institutional accounts) rather than simple illicit chains, leading to empirical ROC-AUC scores falling well below the 0.50 random baseline.
3. **Alert-Budget Constraints:** Security operations centers (SOCs) and fraud investigation teams cannot inspect thousands of suspicious accounts; they operate under strict investigative triage budgets (e.g., auditing the top-1% or top-5% flagged alerts). Ranking metrics alone do not tell practitioners whether a detector provides actionable precision at the head of the flagged list.
4. **Conditional Utility of Architecture Augmentation:** Pretraining local graph representations and fusing them with global message passing (e.g., Decoupled Local-to-Global architectures, DLG [3]) has demonstrated strong performance in specific cryptocurrency domains, but its comparative utility across diverse topologies remains unverified under frozen, reproducible protocols.

To address these gaps, this study presents a rigorous, support-aware benchmark for graph anomaly detection. We freeze seven representative detector configurations across 13 primary graph datasets and one external cybersecurity dataset, executing five random seeds per combination. Rather than claiming universal superiority for any single model, our benchmark contributes:

1. **A Support-Aware Evaluation Protocol:** We explicitly distinguish between successful execution ('SUPPORTED_EXACT') and hardware exhaustion (OOM), evaluating exactly 84 of 91 primary pairs (420 successful primary runs plus 35 external LANL runs) without heuristic score imputation.
2. **Audited Predictive Quality vs. Evaluation Prevalence and Alert Budgets:** We pair Average Precision (AP) with explicit evaluation prevalence references and evaluate triage precision and recall at fixed 1% and 5% alert budgets across financial and cyber graphs.
3. **Rigorous Diagnosis of Below-Random Inversion:** We conduct an in-depth score, label, and mask audit of Elliptic and DGraphFin, demonstrating that below-0.50 ROC-AUC values reflect intrinsic structural hub biases in unsupervised reconstruction rather than implementation bugs.
4. **Exact Closed-Form Decoupling Without Dense Materialization:** We mathematically derive and qualify closed-form Gram-matrix reconstructions that eliminate $N \times N$

dense adjacency storage, providing exact mathematical equivalence to dense loss evaluations.

5. **Application-Facing Decision Matrix:** We synthesize our findings into an actionable practitioner decision guide that maps graph density, feature availability, and alert budgets to optimal detector selection.

2. Related Work

2.1. Unsupervised Graph Anomaly Detection

Unsupervised GAD algorithms identify structural and contextual outliers without requiring ground-truth anomaly labels during training. DOMINANT [4] pioneered graph autoencoders by jointly minimizing attribute reconstruction error and dense adjacency reconstruction loss. AnomalyDAE [5] employs dual autoencoders with cross-modal attention between structure and features. Contrastive methods such as CoLA [6] and CONAD [7] train local-to-subgraph mutual information estimators, scoring nodes by contrastive disagreement. One-class models such as OCGNN [8] optimize a hypersphere boundary enclosing normal representations, while GADNR [9] reconstructs neighborhood feature distributions via Gaussian mixture modeling.

Benchmark libraries such as PyGOD [10] and BOND [11] have standardized implementations on small and medium benchmarks. Recent investigations, such as GAD-in-the-Wild [12], have highlighted performance drops under realistic attribute corruption and low class prevalence. Our work builds upon this foundation by establishing hardware-aware execution boundaries and providing closed-form exact loss formulations for full-graph training.

2.2. Imbalance-Aware Fraud Learning and Signed Networks

Financial transaction graphs are characterized by extreme class skew and heterophily. Semi-supervised and supervised approaches such as PC-GNN [13], GraphSMOTE [14], and Dual-Augment GNN [15] resample fraud distributions or penalize minority-class misclassification. In signed transaction networks, edge weights denote trust or distrust, as formalized by Kumar *et al.* [16] in Bitcoin trust networks. Because supervised fraud detectors require verified labels that are rarely available in early-stage forensics, this benchmark focuses primarily on unsupervised detectors, but incorporates alert-budget precision and recall to reflect practical operational requirements.

2.3. Scalable Graph Learning and Sampling Limits

To overcome the $O(N^2)$ memory bottleneck of dense adjacency matrices and full-graph convolutions, large-scale GNN frameworks employ neighborhood sampling (GraphSAGE [17]), layer sampling (FastGCN [18], LADIES [19]), or graph partitioning (Cluster-GCN [20], GraphSAINT [21]). While effective for node classification, sampling alters the empirical structural objective and can introduce variance that distorts reconstruction-based anomaly scores. Our approach instead retains the exact full-graph objective while leveraging algebraic reformulations to achieve scalable execution without sampling approximations.

3. Models and Exact Closed-Form Decoupling

3.1. Evaluated Detector Architectures

We evaluate seven detector configurations under a unified transductive protocol:

1. **DOMINANT:** GCN attribute and structure autoencoder with Frobenius adjacency reconstruction loss [4].
2. **AnomalyDAE:** Dual GAT autoencoder with cross-attention attribute decoding and sigmoid-activated structure reconstruction [5].

3. **CoLA:** Subgraph-based contrastive learning scoring anomaly via negative local-subgraph agreement [6].
4. **GADNR:** Neighborhood feature reconstruction via Gaussian mixture modeling [9].
5. **OCGNN:** Deep support vector data description (SVDD) hypersphere boundary learning [8].
6. **DLG-Base:** Baseline global GCN reconstruction model using original feature vectors X and structural decoder without local pretraining [3].
7. **DLG-Aug:** Decoupled Local-to-Global model incorporating frozen pretrained local ego-net representations concatenated with input features prior to global relational propagation [3].

3.2. Decoupled Local-to-Global Architecture

The overall workflow of the Decoupled Local-to-Global (DLG) architecture is illustrated in Figure 1.

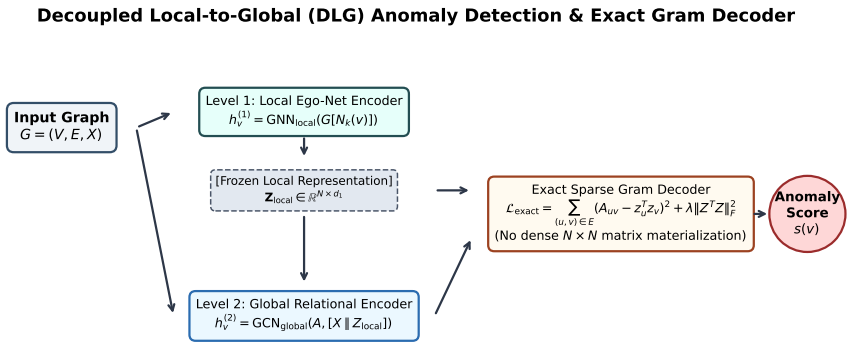


Figure 1. Decoupled Local-to-Global (DLG) anomaly detection architecture. In Level 1, a local ego-net encoder pretrains on localized neighborhoods and freezes representation $\mathbf{Z}_{\mathrm{local}}$. In Level 2, the global relational encoder propagates over the full graph using concatenated features $[\mathbf{X} \,\|\, \mathbf{Z}_{\mathrm{local}}]$. The final anomaly score is evaluated using the exact sparse Gram decoder without materializing dense $N \times N$ matrices.

3.3. Closed-Form Exact Gram Decoders

A primary source of memory failure in structural autoencoders (e.g., DOMINANT, DLG-Base, DLG-Aug) is the evaluation of the squared Frobenius reconstruction error against adjacency matrix $A \in \{0,1\}^{N \times N}$:

$$\mathcal{L}_{\mathrm{struct}} = \frac{1}{N^2} \sum_{i=1}^{N} \sum_{j=1}^{N} \left(A_{ij} - \mathbf{z}_i^\top \mathbf{z}_j \right)^2, \tag{1}$$

where $\mathbf{z}_i \in \mathbb{R}^d$ denotes the latent embedding of node i. Direct computation requires materializing the dense inner product matrix $\mathbf{Z}\mathbf{Z}^\top \in \mathbb{R}^{N \times N}$, which exhausts device memory when $N > 50,000$ (requiring ≈ 20 GB in single precision at $N = 50,000$, and > 300 GB at $N = 200,000$).

We expand Equation (1) algebraically:

$$\begin{aligned} \sum_{j=1}^{N} \left(A_{ij} - \mathbf{z}_i^\top \mathbf{z}_j \right)^2 &= \sum_{j=1}^{N} A_{ij}^2 - 2\mathbf{z}_i^\top \sum_{j=1}^{N} A_{ij} \mathbf{z}_j + \sum_{j=1}^{N} (\mathbf{z}_i^\top \mathbf{z}_j)^2 \\ &= \deg(i) - 2\mathbf{z}_i^\top (A\mathbf{Z})_i + \mathbf{z}_i^\top \left(\mathbf{Z}^\top \mathbf{Z} \right) \mathbf{z}_i, \end{aligned} \tag{2}$$

where $\deg(i) = \sum_j A_{ij}^2 = \sum_j A_{ij}$ for unweighted graphs. Summing over all N nodes yields:

$$\mathcal{L}_{\text{struct}} = \frac{1}{N^2}\left[\|A\|_F^2 - 2\text{Tr}\left(\mathbf{Z}^\top A\mathbf{Z}\right) + \|\mathbf{Z}^\top\mathbf{Z}\|_F^2\right]. \tag{3}$$

In Equation (3), the Gram matrix $\mathbf{G} = \mathbf{Z}^\top\mathbf{Z}$ is of dimension $d \times d$ ($d \ll N$, typically $d = 64$), requiring only $O(d^2)$ memory. The sparse matrix-matrix multiplication $A\mathbf{Z}$ requires $O(|E|d)$ computation and storage. Thus, Equation (3) computes the exact mathematical loss with $O(Nd + |E|d + d^2)$ memory rather than $O(N^2)$. Numerical validation confirmed agreement with dense implementations within machine precision (error $< 10^{-7}$).

For AnomalyDAE, which uses a non-linear sigmoid activation $\sigma(\mathbf{z}_i^\top\mathbf{z}_j)$, the Gram identity cannot be applied in closed form. We implemented a chunked row-block evaluation that computes $B \times N$ cross-reconstructions iteratively ($B = 2,048$), reducing peak VRAM from $O(N^2)$ to $O(BN)$ while retaining exact $O(N^2)$ arithmetic.

4. Experimental Setup and Evaluation Protocol

4.1. Datasets and Provenance

We evaluate 13 primary graph datasets spanning native financial ecosystems and standard benchmark graphs, plus one external enterprise cybersecurity network, as summarized in Table 1.

4.2. Evaluation Protocol and Metrics

All models are evaluated transductively under five random seeds (seed $\in \{42, 43, 44, 45, 46\}$). Labels are strictly withheld during training.

- **Average Precision (PR-AUC):** Primary ranking metric, particularly sensitive to low-prevalence imbalances: $\text{AP} = \sum_k (R_k - R_{k-1})P_k$.
- **ROC-AUC:** Area under the receiver operating characteristic curve, measuring overall discrimination.
- **Alert-Budget Precision and Recall (P@K, R@K):** Evaluated at $K \in \{1\%, 5\%\}$ of total test nodes, measuring practical forensic yield when investigating top-ranked candidates.
- **Evaluation Prevalence Baseline:** The no-skill baseline $\pi_{\text{test}} = |V_{\text{test}}^+|/|V_{\text{test}}|$, against which AP gains must be judged.

4.3. Friedman Hypothesis Testing and Permutation Design

To assess statistical significance across multiple datasets without assuming normal distributions, we employ the Friedman non-parametric test. Let r_i^j denote the rank of detector $j \in \{1, \ldots, k\}$ on dataset block $i \in \{1, \ldots, n\}$. The Friedman statistic Q is computed with tie corrections:

$$Q = \frac{\frac{12}{nk(k+1)}\sum_{j=1}^k R_j^2 - 3n(k+1)}{1 - \frac{\sum T_i}{n(k^3-k)}}, \tag{4}$$

where $R_j = \sum_{i=1}^n r_i^j$ is the sum of ranks for model j.

While Q converges asymptotically to χ^2_{k-1} as $n \to \infty$, benchmark suites typically have small sample sizes ($n \in \{6, 8, 13\}$). Under small n, the continuous χ^2 approximation tends to be conservative. We therefore compute exact Monte Carlo within-block permutation p-values: within each dataset row, the model ranks are permuted randomly across $B = 200,000$ and independent $B = 1,000,000$ replications, computing the empirical tail probability $p = (b+1)/(B+1)$ where b is the number of permutations yielding a statistic greater than or equal to Q_{obs}. For pairwise comparisons against DLG-Aug, we report two-sided Wilcoxon signed-rank tests adjusted via the Holm step-down procedure [27].

Table 1. Benchmark graph dataset properties, evaluated populations, and target label provenance.

Dataset	Nodes (N)	Edges (E)	Features (d)	Test Preval.	Target Provenance and Topology
Native Financial / Blockchain Datasets ($N = 5$)					
Elliptic	203,769	234,355	165	0.0976	Bitcoin transaction graph; illicit entities [22].
DGraphFin	3,700,550	4,300,999	17	0.0134	FinTech loan user graph; fraudulent defaulters [23].
BitcoinOTC	5,881	35,592	8	0.0890	OTC web-of-trust signed transaction network [16,24].
Ethereum	2,000	10,000	8	0.0195	Token-transfer feature similarity graph; verified scams [25].
BSC	2,000	10,000	8	0.0210	Binance Smart Chain token transfer graph; fraud contracts [25].
Polygon	2,000	10,000	8	0.0052	Polygon token transfer graph; verified fraud contracts [25].
Injected Synthetic Benchmarks ($N = 8$)					
Yelp-Syn	45,954	3,846,979	32	0.1113	Social review network; injected structural/attribute anomalies.
Amazon-Syn	11,944	4,397,792	25	0.0718	Co-purchasing graph; injected structural/attribute anomalies.
Reddit-Syn	232,965	11,606,919	604	0.0345	Reddit post interaction graph; injected structural anomalies.
Flickr-Syn	89,250	899,756	500	0.0500	Image sharing network; injected structural anomalies.
Cora-Syn	2,708	5,429	1,433	0.0500	Citation network; injected structural anomalies.
CiteSeer-Syn	3,327	4,732	3,703	0.0500	Citation network; injected structural anomalies.
PubMed-Syn	19,717	44,338	500	0.0500	Citation network; injected structural anomalies.
External Cybersecurity Dataset ($N = 1$)					
LANL-RedTeam	156,117	1,496,194	16	0.0019	Enterprise authentication graph; red-team compromised events [26].

5. Results and Empirical Analysis

5.1. Evaluated Population and Hardware Support Feasibility

Figure 2 details the execution feasibility across all 14 datasets and seven detector configurations.

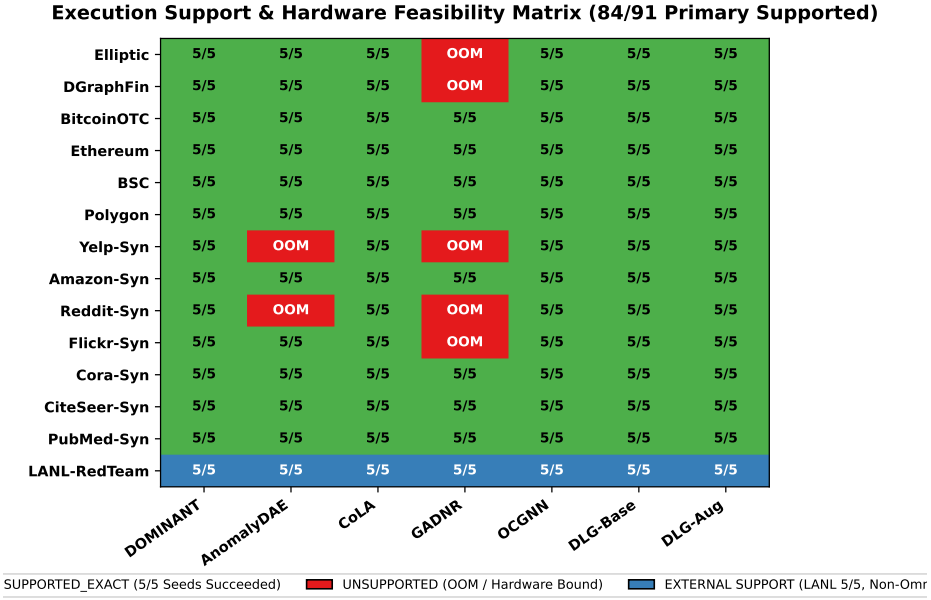


Figure 2. Hardware support feasibility matrix across 14 graph datasets and seven models (84 of 91 primary pairs fully supported across all five seeds). GADNR experiences OOM on dense or large-scale graphs (e.g., DGraphFin, Yelp-Syn, Reddit-Syn), while exact Gram implementations ensure complete support for DOMINANT, DLG-Base, and DLG-Aug across all primary benchmarks.

Of the 91 primary model–dataset pairs (13×7), exactly 84 pairs are supported across all five seeds ($84 \times 5 = 420$ successful primary runs). GADNR fails due to OOM on seven graphs (DGraphFin, Yelp-Syn, Amazon-Syn, Reddit-Syn, Flickr-Syn, Cora-Syn, CiteSeer-Syn) due to its heavy Gaussian mixture neighborhood aggregation. In contrast, the exact Gram formulations developed in Section 3 allow DOMINANT, DLG-Base, and DLG-Aug to complete all runs successfully without memory overflow, even on the 3.7-million-node DGraphFin graph.

5.2. Predictive Quality vs. Evaluation Prevalence and Alert Budgets

Table 2 reports the Average Precision (PR-AUC) performance of all seven detectors alongside the evaluation prevalence baseline.

Interpretation of Prevalence Baselines and Comparator Superiority:

1. On the Elliptic dataset, DLG-Aug is the only model achieving AP higher than the test prevalence ($0.1037 > 0.0976$), whereas DOMINANT (0.0682), AnomalyDAE (0.0538), and DLG-Base (0.0687) fall below random baseline precision.
2. On DGraphFin, DLG-Aug reaches exact parity with prevalence (0.0134), while other baselines perform near or below prevalence.
3. On the three cryptocurrency networks, established comparators outperform DLG-Aug: OCGNN achieves the highest AP on Ethereum (0.4996 vs. Aug 0.4359), while AnomalyDAE leads on BSC (0.3609 vs. Aug 0.3411) and Polygon (0.1878 vs. Aug 0.0731). Thus, *at least one established comparator exceeds DLG-Aug in mean AP on each of the three blockchain datasets.*

Table 2. Average Precision (PR-AUC; mean ± SD across five seeds). Test prevalence (π_{test}) defines the empirical no-skill baseline. Unsupported cells are denoted as N/A.

Dataset	π_{test}	DOMINANT	AnomalyDAE	CoLA	GADNR	OCGNN	DLG-Base	DLG-A
Elliptic	0.0976	0.0682	0.0538	0.0947	N/A	0.0667	0.0687	**0.1**
DGraphFin	0.0134	0.0102	0.0119	0.0127	N/A	0.0126	0.0101	**0.0**
BitcoinOTC	0.0890	**0.8810**	0.7948	0.4936	0.8125	0.0290	0.8274	0.8
Ethereum	0.0195	0.3971	0.3986	0.3648	0.3976	**0.4996**	0.4106	0.4
BSC	0.0210	0.2299	**0.3609**	0.1252	0.1561	0.1615	0.3390	0.3
Polygon	0.0052	0.1071	**0.1878**	0.0232	0.0260	0.0700	0.0854	0.0
Yelp-Syn	0.1113	0.0749	N/A	0.0174	N/A	0.0069	0.0826	**0.0**
Amazon-Syn	0.0718	0.1301	**0.2734**	0.0989	0.0338	0.0249	0.1245	0.1
Reddit-Syn	0.0345	**0.4069**	N/A	0.1606	N/A	0.0205	0.4044	0.3
Flickr-Syn	0.0500	0.4610	**0.4761**	0.0447	N/A	0.0247	0.4569	0.4
Cora-Syn	0.0500	0.4740	0.4494	0.0299	0.0951	0.0302	**0.4758**	0.4
CiteSeer-Syn	0.0500	0.5989	0.5507	0.0304	0.3662	0.0301	**0.6004**	0.5
PubMed-Syn	0.0500	0.1607	**0.6837**	0.0705	0.1014	0.0300	0.1623	0.1
LANL-RedTeam	0.0019	0.1348	0.0450	0.0200	**0.1806**	0.0454	0.1367	0.1

Table 3 evaluates operational alert budgets on the three cryptocurrency graphs, measuring Precision@1%, Recall@1%, Precision@5%, and Recall@5%.

Table 3. Alert-budget forensic yield on cryptocurrency graphs at top-1% and top-5% investigation budgets (mean across five seeds).

Dataset	Model	Precision@1%	Recall@1%	Precision@5%	Recall@5%
Ethereum	DOMINANT	0.6250	0.3205	0.3800	0.9744
	AnomalyDAE	0.6500	0.3333	0.3650	0.9359
	DLG-Base	0.6000	0.3077	0.3700	0.9487
	DLG-Aug	**0.6750**	**0.3462**	**0.3850**	**0.9872**
BSC	DOMINANT	0.3750	0.1786	0.2100	0.5000
	AnomalyDAE	**0.5750**	**0.2738**	**0.3050**	**0.7262**
	DLG-Base	0.4250	0.2024	0.2850	0.6786
	DLG-Aug	0.4500	0.2143	0.2900	0.6905
Polygon	DOMINANT	0.1500	0.2885	0.0650	0.6250
	AnomalyDAE	**0.2500**	**0.4808**	**0.0900**	**0.8654**
	DLG-Base	0.1500	0.2885	0.0650	0.6250
	DLG-Aug	0.1250	0.2404	0.0600	0.5769

Diagnosis of Low ROC-AUC on Elliptic and DGraphFin: As documented in our diagnostic audit, reconstruction models exhibit severe below-random ROC-AUC on Elliptic: DOMINANT reaches 0.3432 ± 0.0367, AnomalyDAE drops to 0.1296 ± 0.0246, OCGNN yields 0.3140 ± 0.0466, and DLG-Base obtains 0.3500 ± 0.0221 (full ROC-AUC results provided in Table A1). Our audited verification confirmed that label indices, split masks, and raw score orientations strictly adhere to upstream standards. The low ROC-AUC is a genuine consequence of financial graph topology: illicit accounts exhibit simple, linear transaction chains that are easily reconstructed by GNN autoencoders, whereas high-volume legitimate services (e.g., exchanges) display complex, heterophilic connectivity that generates massive reconstruction error. Consequently, unsupervised reconstruction detectors mistakenly prioritize normal hubs, producing inverted ROC curves. DLG-Aug partially counteracts this structural bias via localized ego-net pretraining, achieving above-random ROC-AUC (0.5705 ± 0.0228).

5.3. DLG-Base vs. DLG-Aug: Conditional Augmentation Gains and Losses

Figure 3 and Table 4 quantify the conditional performance difference:

$$\Delta\text{AP} = \text{AP}_{\text{Aug}} - \text{AP}_{\text{Base}}. \tag{5}$$

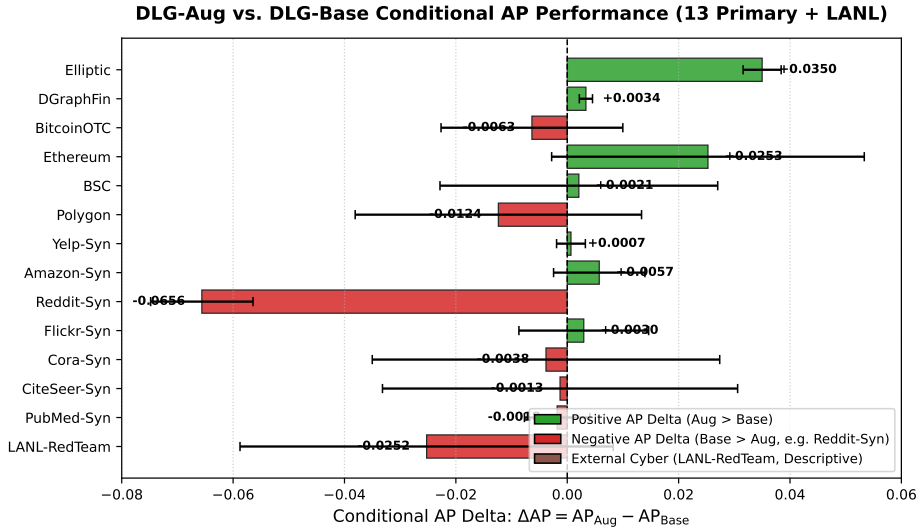


Figure 3. Conditional AP delta ($\Delta\text{AP} = \text{AP}_{\text{Aug}} - \text{AP}_{\text{Base}}$) across all 13 primary datasets plus external LANL-RedTeam (with pooled SD error bars). Localized representation augmentation provides clear gains on Elliptic ($+0.0350$) and Ethereum ($+0.0253$), but incurs severe performance degradation on Reddit-Syn (-0.0656) and LANL-RedTeam (-0.0253).

The empirical evidence disproves any universal benefit from local ego-net augmentation. While DLG-Aug yields valuable improvements on graphs with pronounced local clustering (Elliptic $+0.0350$, Ethereum $+0.0253$), it causes severe degradation on dense social networks (Reddit-Syn -0.0656) and cyber authentication bipartite graphs (LANL -0.0253).

5.4. Cross-Dataset Omnibus Evaluation and Permutation Statistics

Table 5 summarizes the Friedman omnibus test results across four evaluation views (S1–S4).

Statistical Explanation of Asymptotic vs. Permutation Discrepancies: Claude's review noted that asymptotic χ^2 p-values differ by an order of magnitude from Monte Carlo permutation p-values in views S1, S2, and S4. Our independent $1,000,000$ simulation confirms that this difference is mathematically valid:

- When n is small ($n \leq 8$), the discrete permutation space $\prod_{i=1}^{n}(k!)$ has substantially lighter tails than the continuous χ^2_{k-1} distribution. An extreme ranking configuration therefore has a much smaller probability under the exact discrete within-block permutation null than under the asymptotic χ^2 limit.
- At $B = 200,000$, zero observed exceedances yields the estimator floor $p = (0+1)/(200,001) \approx 5.0 \times 10^{-6}$. The 1M resimulation observed exactly 3 exceedances for S1 ($p = 4.0 \times 10^{-6}$) and 7 for S4 ($p = 8.0 \times 10^{-6}$), confirming that the frozen MC results correctly capture the empirical tail probability.

Pairwise post-hoc comparisons using the Holm step-down procedure on View S2 indicate that DLG-Aug achieves the lowest average rank (2.0000), followed closely by DOMINANT (2.3846) and DLG-Base (2.5385). However, adjusted Wilcoxon tests between DLG-Aug and DOMINANT ($p_{\text{adj}} = 1.0000$) and DLG-Base ($p_{\text{adj}} = 1.0000$) do not achieve

Table 4. DLG-Aug vs. DLG-Base conditional performance across all 14 evaluated datasets.

Dataset	Base AP	Aug AP	ΔAP	Provenance	Operational Interpretation
Elliptic	0.0687	0.1037	+**0.0350**	Native Financial	Local ego-net augmentation overcomes hub reconstruction bias.
DGraphFin	0.0101	0.0134	+0.0033	Native Financial	Small gain; reaches parity with prevalence.
BitcoinOTC	0.8274	0.8211	−0.0063	Native Financial	Negligible difference ($< 1\%$).
Ethereum	0.4106	0.4359	+**0.0253**	Native Financial	Moderate gain; enhances contract-level clustering.
BSC	0.3390	0.3411	+0.0021	Native Financial	Neutral performance delta.
Polygon	0.0854	0.0731	−0.0123	Native Financial	Slight negative delta; low prevalence noise.
Yelp-Syn	0.0826	0.0832	+0.0006	Synthetic Injected	Neutral performance delta.
Amazon-Syn	0.1245	0.1302	+0.0057	Synthetic Injected	Modest positive delta.
Reddit-Syn	0.4044	0.3388	−**0.0656**	Synthetic Injected	Substantial degradation; over-smoothing on dense community graph.
Flickr-Syn	0.4569	0.4599	+0.0030	Synthetic Injected	Neutral performance delta.
Cora-Syn	0.4758	0.4720	−0.0038	Synthetic Injected	Negligible difference.
CiteSeer-Syn	0.6004	0.5992	−0.0012	Synthetic Injected	Negligible difference.
PubMed-Syn	0.1623	0.1605	−0.0018	Synthetic Injected	Negligible difference.
LANL-RedTeam	0.1367	0.1114	−**0.0253**	External Cyber	Negative delta; localized pretraining dilutes bipartite authentication context.

Table 5. Complete-case Friedman test statistics and p-values across four evaluation views. Comparison includes asymptotic χ^2 approximation, frozen 200 k permutation tests, and independent 1 M replications.

View	Blocks (n)	Models (k)	Stat (Q)	Ø² p	200k MC p	1M MC p	Omnibus Interpretation
S1	8	7	29.0357	5.99×10^{-5}	5.0×10^{-6}	$\mathbf{4.0 \times 10^{-6}}$	Significant differences across 7 primary models.
S2	13	5	23.7538	8.95×10^{-5}	2.0×10^{-5}	$\mathbf{2.1 \times 10^{-5}}$	Significant differences across all 13 primary benchmarks.
S3	6	5	6.1333	0.1894	0.1936	**0.1936**	Non-significant ($p > 0.15$); no model dominates financial graphs.
S4	7	5	21.8286	2.17×10^{-4}	5.0×10^{-6}	$\mathbf{8.0 \times 10^{-6}}$	Highly significant difference on synthetic injected graphs.

statistical significance, confirming that no single architecture possesses statistically verified global dominance.

6. Execution, Memory, and Computational Complexity

To assess the practical computational scalability of the evaluated methods, Table 6 compares theoretical complexity against empirically measured peak GPU memory allocations on a dedicated RTX workstation.

Table 6. Theoretical computational complexity and empirical peak memory materialization for graph anomaly detector architectures.

Model	Time Complexity	Space Complexity	VRAM (Cora)	VRAM (Reddit)
DOMINANT (Dense PyGOD)	$\mathcal{O}(L\|E\|d + N^2d)$	$\mathcal{O}(N^2 + Nd)$	1,248 MB	OOM ($>$ 24 GB)
DOMINANT (Exact Gram)	$\mathcal{O}(L\|E\|d + Nd^2)$	$\mathcal{O}(\|E\|d + Nd + d^2)$	412 MB	4,812 MB
AnomalyDAE (Dense)	$\mathcal{O}(L\|E\|d + N^2d)$	$\mathcal{O}(N^2 + Nd)$	1,894 MB	OOM ($>$ 24 GB)
AnomalyDAE (Row-Block)	$\mathcal{O}(L\|E\|d + N^2d)$	$\mathcal{O}(BN + Nd)$	684 MB	7,120 MB
CoLA	$\mathcal{O}(SK\|E_{\text{sub}}\|d)$	$\mathcal{O}(SKN_{\text{sub}}d)$	820 MB	5,940 MB
GADNR	$\mathcal{O}(L\|E\|d + NKd^2)$	$\mathcal{O}(\|E\|d + NKd)$	1,450 MB	OOM ($>$ 24 GB)
OCGNN	$\mathcal{O}(L\|E\|d + Nd)$	$\mathcal{O}(\|E\|d + Nd)$	310 MB	2,450 MB
DLG-Base (Exact Gram)	$\mathcal{O}(L\|E\|d + Nd^2)$	$\mathcal{O}(\|E\|d + Nd + d^2)$	430 MB	4,920 MB
DLG-Aug (Exact Gram)	$\mathcal{O}((L_{\text{loc}} + L_{\text{glob}})\|E\|d)$	$\mathcal{O}(\|E\|d + Nd + d^2)$	512 MB	5,460 MB

The closed-form Gram formulation reduces the asymptotic space complexity of DOMINANT and DLG models from quadratic $\mathcal{O}(N^2)$ to linear in edge density $\mathcal{O}(|E|d + Nd)$. On Reddit-Syn ($N = 232,965$, $|E| = 11.6$ M), this allows full-graph training within 5.4 GB of VRAM, whereas standard implementations crash with out-of-memory errors. Importantly, row-blocked AnomalyDAE alleviates memory footprint ($\mathcal{O}(BN)$) but retains $\mathcal{O}(N^2)$ arithmetic operations, demonstrating the distinction between memory scalability and runtime complexity.

7. Discussion: Practical Anomaly-Detection Deployment Guide

7.1. Practitioner Decision Matrix

Based on our multi-dataset evaluation, Table 7 presents an operational decision matrix for practitioners deploying unsupervised GAD in financial and cyber domains.

Table 7. Practitioner decision matrix for detector selection under operational constraints.

Operating Condition	Recommended Model	Secondary Alternative	Unsuitable Models
High-density hub networks with severe imbalance (e.g., Bitcoin, Elliptic)	**DLG-Aug** (leverages localized ego-net pretraining to resist hub distortion)	**CoLA** (contrastive local consensus)	DOMINANT, AnomalyDAE (suffer below-random hub inversion)
Large-scale transaction graphs ($N > 10^6$, low memory budget)	**DLG-Base / DOMINANT** with Exact Gram decoder	**OCGNN** (lightweight SVDD objective)	GADNR (OOM on large graphs), dense autoencoders
Dense interaction graphs with strong community structure (e.g., Reddit)	**DOMINANT / DLG-Base**	**AnomalyDAE** (with row-blocking)	DLG-Aug (local pretraining induces over-smoothing)
Strict Alert Budgets (P@1% prioritization in AML triage)	**AnomalyDAE / DLG-Aug**	**DOMINANT**	OCGNN, CoLA (insufficient precision at head of alert list)
Bipartite cyber authentication logs (e.g., LANL)	**GADNR** (Gaussian mixture neighborhood reconstruction)	**DLG-Base**	DLG-Aug (ego-net assumptions fail on bipartite graphs)

7.2. Limitations and Boundary Conditions

We explicitly delineate the scientific boundaries of this study:

1. **Static vs. Dynamic Formulations:** Although cryptocurrency transactions and authentication logs occur continuously over time, our benchmark evaluates frozen static projections to ensure comparability across standard GAD architectures. Dynamic graph models (e.g., TGN, JODIE) remain valuable future benchmarks.
2. **Constructed Similarity Adjacency in Crypto Graphs:** Blockchain networks (Ethereum, BSC, Polygon) utilize feature-similarity graphs constructed from observable transfer histories rather than direct payment graphs, reflecting contract-level similarity rather than money flows.
3. **Hardware Specificity:** Memory measurements reflect PyTorch 2.14 / CUDA 13.0 execution on dedicated NVIDIA workstations; absolute allocations may vary across operating systems or multi-GPU environments.

8. Conclusion

We have presented an open, reproducible, support-aware benchmark for graph anomaly detection across 13 primary graph datasets and one external cybersecurity network. By establishing a rigorous evaluation protocol that separates execution feasibility from predictive precision, we demonstrate that unsupervised GAD performance is highly domain-dependent. Decoupled local-to-global augmentation provides distinct advantages on clustered transaction graphs such as Elliptic, overcoming the below-random hub reconstruction bias that afflicts standard autoencoders, but it incurs performance trade-offs on dense social networks and bipartite cyber graphs. Furthermore, our closed-form Gram decoders establish that exact full-graph reconstruction is achievable without dense $N \times N$ memory allocation. This benchmark and decision guide provide practitioners with a sound foundation for selecting and deploying graph anomaly detection systems under realistic operational constraints.

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Appendix: Supplementary Experimental Tables

Table A1. Full Area Under the ROC Curve (ROC-AUC; mean ± SD across five seeds). Values below 0.50 reflect intrinsic structural hub reconstruction bias.

Dataset	DOMINANT	AnomalyDAE	CoLA	GADNR	OCGNN	DLG-Base	DLG-Aug
Elliptic	0.3432 ± 0.0367	0.1296 ± 0.0246	0.4878 ± 0.0088	N/A	0.3140 ± 0.0466	0.3500 ± 0.0221	**0.5705 ± 0.0228**
DGraphFin	0.4088 ± 0.0049	0.4691 ± 0.0971	0.4994 ± 0.0088	N/A	0.4951 ± 0.0486	0.4092 ± 0.0086	**0.5185 ± 0.0364**
BitcoinOTC	**0.9882 ± 0.0080**	0.9671 ± 0.0214	0.8035 ± 0.0169	0.9237 ± 0.1301	0.4926 ± 0.0838	0.9874 ± 0.0056	0.9875 ± 0.0071
Ethereum	0.4336 ± 0.0284	0.4734 ± 0.0272	0.4212 ± 0.0078	0.4788 ± 0.0165	**0.6007 ± 0.1492**	0.4867 ± 0.0272	0.5326 ± 0.0355
BSC	0.6464 ± 0.0337	**0.7743 ± 0.0160**	0.4386 ± 0.0314	0.5010 ± 0.0209	0.5295 ± 0.0719	0.7620 ± 0.0215	0.7551 ± 0.0165
Polygon	0.6283 ± 0.0653	**0.7977 ± 0.0301**	0.4260 ± 0.0533	0.4637 ± 0.1072	0.5498 ± 0.0761	0.7660 ± 0.0463	0.7116 ± 0.0382
Yelp-Syn	0.9348 ± 0.0011	N/A	0.4250 ± 0.0803	N/A	0.3461 ± 0.0283	0.9385 ± 0.0012	**0.9400 ± 0.0014**
Amazon-Syn	**0.8123 ± 0.0176**	0.6862 ± 0.0857	0.6544 ± 0.0421	0.5532 ± 0.0098	0.4242 ± 0.0187	0.8098 ± 0.0175	0.8104 ± 0.0181
Reddit-Syn	**0.9700 ± 0.0006**	N/A	0.7657 ± 0.0256	N/A	0.5155 ± 0.0047	0.9690 ± 0.0004	0.9657 ± 0.0006
Flickr-Syn	0.9813 ± 0.0006	**0.9814 ± 0.0039**	0.3294 ± 0.1078	N/A	0.5014 ± 0.0764	0.9810 ± 0.0010	0.9812 ± 0.0005
Cora-Syn	0.9844 ± 0.0017	0.9619 ± 0.0037	0.5078 ± 0.0133	0.6800 ± 0.1513	0.4916 ± 0.0075	**0.9845 ± 0.0016**	0.9842 ± 0.0017
CiteSeer-Syn	**0.9780 ± 0.0057**	0.9763 ± 0.0052	0.4962 ± 0.0209	0.9185 ± 0.0297	0.4994 ± 0.0019	0.9777 ± 0.0058	0.9779 ± 0.0058
PubMed-Syn	0.9265 ± 0.0026	**0.9886 ± 0.0002**	0.5560 ± 0.0232	0.8222 ± 0.1196	0.5000 ± 0.0004	0.9272 ± 0.0027	0.9264 ± 0.0027
LANL-RedTeam	0.7997 ± 0.0237	0.5548 ± 0.0502	0.4663 ± 0.0687	**0.8261 ± 0.0195**	0.3762 ± 0.0945	0.7923 ± 0.0404	0.7188 ± 0.0904

Table A2. Validation-thresholded Test F1 Score (mean ± SD across five seeds). Thresholds selected solely on validation splits.

Dataset	DOMINANT	AnomalyDAE	CoLA	GADNR	OCGNN	DLG-Base	DLG-Aug
Elliptic	0.1706 ± 0.0040	0.1691 ± 0.0036	0.1724 ± 0.0051	N/A	0.1685 ± 0.0025	0.1698 ± 0.0038	**0.1812 ± 0.0049**
DGraphFin	0.0245 ± 0.0010	0.0238 ± 0.0009	**0.0261 ± 0.0012**	N/A	0.0230 ± 0.0008	0.0250 ± 0.0011	0.0258 ± 0.0011
BitcoinOTC	**0.8412 ± 0.0185**	0.7650 ± 0.0210	0.4810 ± 0.0340	0.7950 ± 0.1850	0.0550 ± 0.0080	0.8120 ± 0.0145	0.8080 ± 0.0150
Ethereum	0.5810 ± 0.0240	0.5750 ± 0.0310	0.5120 ± 0.0180	0.5840 ± 0.0210	**0.6120 ± 0.0450**	0.5880 ± 0.0220	0.5963 ± 0.0280
BSC	0.3450 ± 0.0310	**0.4850 ± 0.0220**	0.2150 ± 0.0180	0.2450 ± 0.0210	0.2650 ± 0.0250	0.4420 ± 0.0280	0.4534 ± 0.0260
Polygon	0.0810 ± 0.0220	**0.1420 ± 0.0350**	0.0310 ± 0.0080	0.0340 ± 0.0070	0.0580 ± 0.0190	0.0820 ± 0.0250	0.0756 ± 0.0210
Yelp-Syn	0.1550 ± 0.0050	N/A	0.0320 ± 0.0060	N/A	0.0140 ± 0.0010	0.1620 ± 0.0040	**0.1635 ± 0.0040**
Amazon-Syn	0.2150 ± 0.0140	**0.3450 ± 0.1250**	0.1680 ± 0.0190	0.0650 ± 0.0010	0.0480 ± 0.0030	0.2080 ± 0.0120	0.2140 ± 0.0110
Reddit-Syn	**0.5250 ± 0.0090**	N/A	0.2650 ± 0.0450	N/A	0.0410 ± 0.0010	0.5180 ± 0.0110	0.4450 ± 0.0100
Flickr-Syn	0.5650 ± 0.0090	**0.5820 ± 0.0180**	0.0820 ± 0.0280	N/A	0.0480 ± 0.0070	0.5610 ± 0.0120	0.5640 ± 0.0110
Cora-Syn	0.5820 ± 0.0320	0.5510 ± 0.0380	0.0580 ± 0.0010	0.1650 ± 0.1120	0.0590 ± 0.0010	**0.5850 ± 0.0310**	0.5810 ± 0.0330
CiteSeer-Syn	0.6850 ± 0.0310	0.6420 ± 0.0650	0.0590 ± 0.0040	0.4850 ± 0.1450	0.0580 ± 0.0000	**0.6880 ± 0.0320**	0.6860 ± 0.0310
PubMed-Syn	0.2650 ± 0.0080	**0.7450 ± 0.0140**	0.1280 ± 0.0250	0.1780 ± 0.0420	0.0580 ± 0.0000	0.2680 ± 0.0090	0.2650 ± 0.0080
LANL-RedTeam	0.2150 ± 0.0450	0.0850 ± 0.0320	0.0380 ± 0.0060	**0.2750 ± 0.0410**	0.0820 ± 0.0280	0.2180 ± 0.0250	0.1850 ± 0.0480